

A Data-Driven Approach for Enhancement of Propeller Performance Prediction

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Accurate and efficient prediction of propeller aerodynamic and acoustic performance is of paramount importance to the analysis and design of various urban air mobility concepts emerging over the last decade. This paper presents the first exploratory effort in developing a data-driven framework to improve the prediction of propeller performance. In this framework, a deep neural network machine learning (ML) model is trained in a multi-fidelity manner using transfer learning (TL) and active learning (AL). The model is first trained using a large number of computationally inexpensive low-fidelity simulations and then enhanced by a small number of high-fidelity aeroacoustic wind tunnel measurements using TL. Additional propellers which were manufactured and experimentally measured were intelligently selected based on an AL algorithm designed to minimize the predictive error of the ML model at two farfield observer locations. In the deployment stage, this ML model enables rapid prediction of the tonal noise level of isolated 2-bladed propellers at two farfield observer locations given the airfoil profile, pitch-to-diameter ratio as well as forward speed and rotational speed of the propeller. Results evaluated based on experimental data from two extra sets of propeller blades, which were held out from the ML training process, indicated that the multi-fidelity ML model trained with both low- and high-fidelity data based on the TL approach delivered more accurate predictions than the ML models trained solely with low- or high-fidelity data. Furthermore, the inclusion of additional high-fidelity data set selected by the AL algorithm was shown to further improve the predictive accuracy of the ML model.

I. Introduction

Due to the emergence of Urban Air Mobility (UAM) concepts, propeller-powered systems have garnered renewed interest in several sectors such as the aircraft and automotive manufacturers, research institutions and academia, as well as various regulators and government agencies. These vehicles are expected to drastically transform the mode of transport and lifestyles of people in urban areas, with a positive social-economic impact. Typically installed on lighter and slower vehicles, these electric and distributed propulsion systems demand less engine power and attain lower blade tip speed, allowing them to achieve 10-20% higher fuel efficiency over typical turbofan engines in static and flight test conditions.¹ As they are mostly designed to operate in urban environments, a major roadblock to their public acceptance is their noise level. To enable the anticipated wide-spread use of UAM vehicles, aeroacoustic considerations must be the central focus of the design and no longer treated as a side-constraint. However, accurate prediction of propeller aerodynamic and aeroacoustic performance, especially in off-design conditions remains challenging.

Propeller noise is the result of relative motion between the propeller blades and the surrounding air. In particular, it consists of both a tonal and a broadband component. The tonal component is produced by the displacement of air due to the blade motion (thickness noise) and the mean pressure difference between the top and bottom surfaces of the

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blades as they rotate through the air (loading noise). The broadband component, on the other hand, occurs as a result of turbulence in the boundary layer and the wake of the propeller blades.

Accurate aeroacoustic prediction of propellers, especially in the broadband range, requires high-fidelity scale-resolving simulations such as Lattice-Boltzmann Method (LBM)² or large-eddy simulation (LES).³ However, these methods are computationally intensive and therefore not suitable for many-query tasks such as design optimization. On the other hand, low- and mid-fidelity methods such as those based on Blade Element Momentum Theory (BEMT) and vortex-lattice method (VLM) tend to perform poorly under off-design conditions and highly twisted blades, and are incapable of capturing the broadband noise component.

In this work, we seek to circumvent this predicament by exploring a data-driven framework where a predictive machine learning (ML) model is trained using a large number of relatively inexpensive low-fidelity simulations and enhanced by a small high-fidelity data set obtained from wind-tunnel experiments. In the deployment stage, this ML model can be used to predict the acoustic (farfield noise level) performances of isolated propellers in forward flight, and each prediction only requires an evaluation of the ML model which is usually extremely fast.

Specifically, we will use transfer learning (TL)⁴ to incorporate information from multiple fidelity of data sources, and active learning (AL)⁵ to intelligently select the next high-fidelity experiments to perform. The ML model will be initially built using a relatively large data set from inexpensive low-fidelity simulations. This baseline ML model is then “transfer learned” by tuning with a few data points of high-fidelity wind tunnel experiments. We then further improve the overall ML model by conducting additional wind tunnel experiments. Since these experiments are quite expensive, entailing production of propeller blade prototypes and expenses from reserving the wind tunnel facility, we need to be selective and identify the experiment that may offer the greatest improvement to the ML model. We will achieve this through AL, where we build a Gaussian process⁶ on model discrepancy and use it to guide the selection of new experiments in regions with high discrepancy.

For low-fidelity simulations, we use a BEMT solver to predict propeller thrust and torque. The BEMT solver is coupled with the Garrick-Watkins (GW)⁷ isolated propeller noise model which allows the prediction of sound pressure level at blade harmonics at farfield observer locations. The time-to-solution of the BEMT-GW predictions is several orders of magnitude lower than that of high-fidelity scale-resolving simulations and hence allows for a large number of low-fidelity simulations to be performed for the initial ML training. The high-fidelity experiments have been performed in the aeroacoustic wind tunnel at the University of Bristol using 3D printed propeller blades. In this first exploratory study, we opt to focus on enhancing the propeller *tonal noise* prediction using the aforementioned data-driven approach, deferring the prediction of broadband noise to follow-on studies.

The remainder of this paper is organized as follows: various components of the ML-based propeller prediction framework including the low-fidelity simulation method, experimental set-up, transfer and active learning approaches are described in Section II. The results are presented in Section III, while the conclusion and outlook for future work are outlined in Section IV.

II. Methodology

The overall framework of the multi-fidelity data-driven model for propeller noise prediction is developed as follows: the definition of propeller design space and the selection of input features for the machine learning study is first determined in Section II.A. A large number of randomly generated samples within this design space are then simulated using the low-fidelity propeller noise prediction tool presented in Section II.B. The details of the high-fidelity aeroacoustic experiment, including blade design, manufacturing and wind tunnel measurements are discussed in Section II.C.2. The baseline machine learning model trained initially with only the low-fidelity data is enriched by pilot experimental data using the transfer learning technique outlined in Section II.D. To further reduce the predictive error of this multi-fidelity machine learning model, an active learning technique, outlined in Section II.E is applied to intelligently determine the additional propeller blades to be manufactured and their associated experiments performed.

II.A. Definition of Propeller Design Space

For this study, we focus only on two-bladed propellers with a fixed diameter of 22cm. The blade twist and cross-sectional profile are both known to have a strong influence on the aerodynamic performance and noise signature of an isolated propeller and are therefore allowed to vary. Specifically, blade twist was varied via the pitch-to-diameter ratio (P/D), while the cross-sectional profiles of the propeller blades were controlled by the three parameters that define the conventional NACA 4-digit airfoil profiles, namely maximum camber (M_c), location of maximum camber (C_{max}) and thickness (t_c). A constant airfoil profile was used over the full span of the blades. No level of taper was introduced to

any of the blades, with each maintaining a constant c/R of 0.2, corresponding to a constant chord length of $c = 2.3\text{cm}$. Figure 1 shows the blade angle (β) distribution and the chord length along the spanwise direction for a representative propeller case with a $P/D = 0.8$. The tip of each blade was kept square. In addition, for each propeller, the flow speed (U_∞) and rotational speed (Ω) are both considered design variables, while the incoming flow is assumed to be uniform and parallel to the axis of rotation of the propeller.

In summary, the propeller design variables selected for this study consist of four propeller shape variables (M_c , C_{max} , t_c , and P/D) and two operating conditions (Ω and U_∞). These constitutes the six input features of the ML model. Table 1 shows the upper and lower bounds imposed on these design variables. Furthermore, it was observed in preliminary experimental studies that as the flow speeds is increased above 10m/s, the propeller tends to start windmilling at lower RPMs. To avoid this, the design space is further restricted as shown in Figure 2.

The two output quantities of interest ($SPL@45^\circ$ and $SPL@90^\circ$) from the ML model were defined to be the sound pressure level (SPL) at the blade passing frequency at the 45° and 90° microphones located at a distance of 1.75m from the center of the propeller hub.

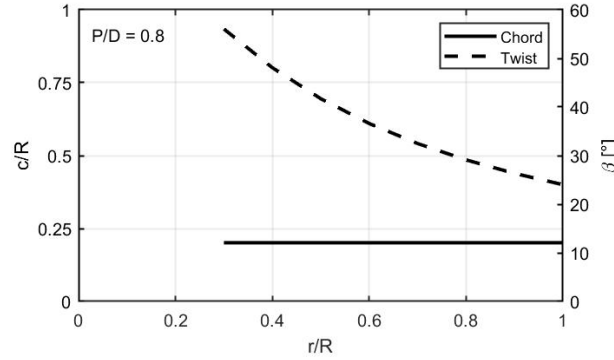


Figure 1: Representative spanwise angle distributions and chord lengths for the blade with $P/D = 0.8$.

Design Variable		Lower Bound	Upper Bound
Maximum Camber (% chord)	M_c	0	5.0
Location of Maximum Camber (% chord)	C_{max}	20.0	50.0
Thickness (% chord)	t_c	10.0	20.0
Pitch-to-Diameter Ratio	P/D	0.8	1.4
Flow Speed (m/s)	U_∞	8.0	20.0
Rotational Speed (RPM)	Ω	3000	6000

Table 1: Propeller design variables and their bounds considered in this study.

II.B. Low-Fidelity Propeller Simulation

II.B.1. Propeller Load Simulation with BEMT

For predicting the load produced by a propeller in forward flight, Blade Element Momentum Theory (BEMT) is used. BEMT is a combination of two widely known analytical methods for estimating propeller loads: blade element theory and momentum theory. Blade element theory splits the propeller blade along the radial direction into a number of 2D sections. The forces on each section can then be obtained through aerofoil theory and integrated along the blade to produce the propeller thrust and torque. This method is combined with momentum theory, which treats the propeller as an actuator disk. By combining these two methods the induction of the freestream fluid by the propeller into the rotor plane is taken into account when estimating the 2D sectional loads, producing a more accurate prediction. Spanwise flow effects along the blade are not modelled by BEMT, so a Prandtl tip loss function is included in the implementation to better predict the reduction in sectional lift near the tip and hub. The aerofoil load estimation is performed using a deep neural network that was trained using a large set of XFOIL predictions. More details on this BEMT implementation are provided by Pullin *et al.*⁸

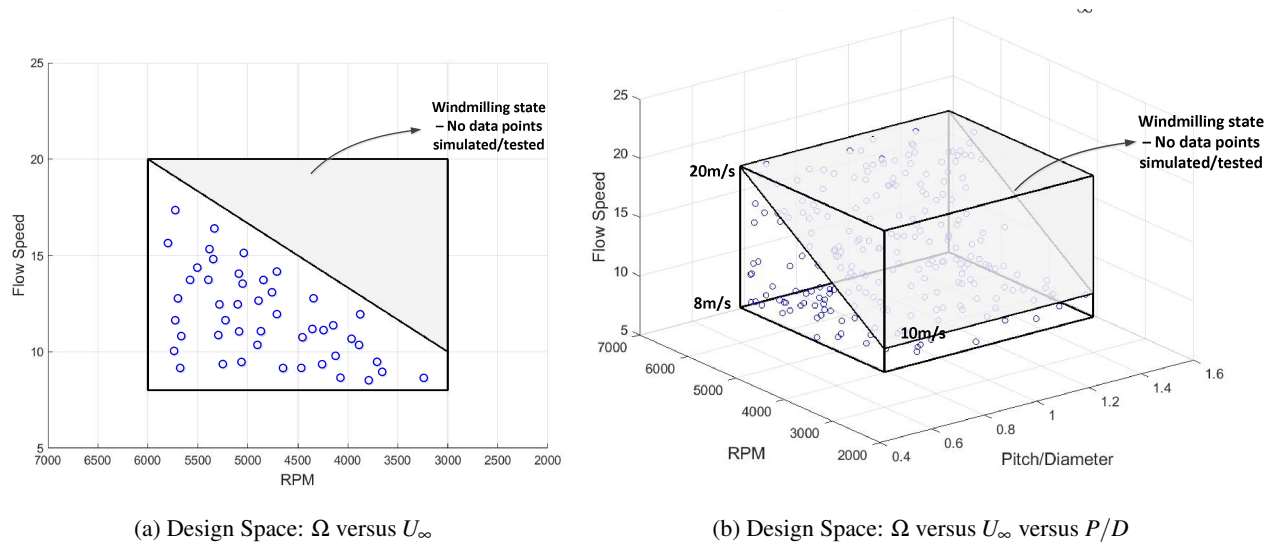


Figure 2: Further restriction of propeller design space to avoid wind-milling (grey region).

II.B.2. Noise Prediction using the Garrick-Watkins Method

The acoustic prediction for the tone generated by an isolated propeller is achieved using a method derived by Garrick and Watkins.⁷ This method assumes that an infinitesimal element of the rotor disk imparts a force on the fluid surrounding it that is harmonic in nature. The impact of this force on the surrounding fluid produces a pressure fluctuation which can be described as an acoustic source. This element is then integrated over an annular ring at which the loads on the rotor disk are assumed to act. The components of the acoustic pressure field generated by the thrust and torque on the rotor can then be found and evaluated at an observer location, which can be used to evaluate acoustic metrics. Further details on this method can be found in the original paper. Only a prediction for the tonal noise based on the blade loading is possible through this method. Other tonal sources of noise such as blade-vortex interaction are neglected and broadband noise generation is not included for this study. For the acoustic results presented in this paper only the fundamental harmonic at the blade passing frequency is considered, as it is generally the most dominant tonal noise source.

II.C. Experimental Set-up

II.C.1. Blade Design and Manufacturing

Since there are a number of parameters influencing the aerodynamic performance and noise signature of an isolated propeller, such as the cross-sectional aerofoil profile, blade twist, pitch and taper angles, as well as tip geometry, the propellers used in the present study are designed in-house to allow full control of these parameters. As a proof-of-concept study, parametric variations of the propeller blades are limited to the aerofoil profiles and the pitch angles.

The propeller blades were manufactured using additive manufacturing techniques with a resin based stereolithography apparatus (SLA) printer, which has been previously verified to produce propellers with accurate profile resolution, sufficiently smooth surface finish and material strength at moderate rotational speed. The SLA prints were oriented such that the layers of plastic were printed in the radial direction towards the tip of each blade, achieving an improved surface finish with minimal support material. The absence of substantial support material reduced the time required to post-process each blade. By printing in the radial direction each propeller blade maintained a smooth airfoil profile throughout its length. The blades were then mated with a hub manufactured by varioPROP,⁹ making it simple to assemble into a two-bladed propeller. An image of one of the SLA-printed blades fitted to the varioPROP hub is shown in Figure 3.

II.C.2. Experimental Condition and Measurements

To obtain high-fidelity propeller performance data necessary for the ML training, the experiments were conducted at the aeroacoustic wind tunnel facility at University of Bristol. The open jet has a nozzle dimension of 0.775m (H) ×

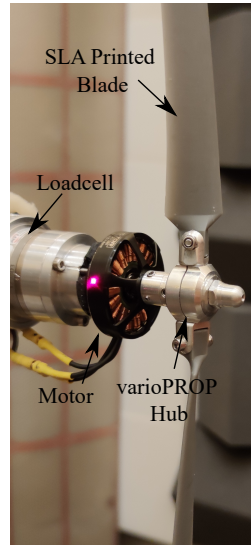


Figure 3: Photograph of one of the SLA blades fitted to the varioPROP hub on the experimental test setup. The motor and loadcell are also labelled.

0.5m (W), corresponding to a blockage factor of less than 10%. More details of the wind tunnel facility can be found in Mayer et al.¹⁰ The experiments were carried out at four free-stream velocities from $U_\infty = 8\text{m/s}$ to 20m/s at 4m/s interval and the rotational speed of the propeller was controlled through an electronic speed controller (ESC) to vary from 3000rpm (revolution-per-minute) to 6000rpm to examine the effects of advance ratios on the noise signature. Here, the advance ratio, J , is defined as $J = U_\infty / (nD)$, where n is the rotational speed in revolution per second.

Figure 4 provides an overview of the propeller set-up of the present experiment. The far-field noise level and directivity were measured from the overhead microphone array, consisting of 23 GRAS 40PL free-field microphones spanning a polar angle range from 40° to 150° . Information from all 23 microphones were collected simultaneously using National Instruments sound acquisition system for a duration of 16s at sampling rate of 2^{16}Hz . The collected data were then subjected to Welch's method in Matlab to transform into frequency domain and the power spectral density of the far-field noise was determined as $10\log_{10}(\phi_{pp}/p_0^2)$, where ϕ_{pp} is magnitude squared unsteady pressure fluctuation and p_0 denotes the reference pressure of $2 \times 10^{-5}\text{Pa}$. The frequency bin (δf) of the Welch's analysis was kept at 1Hz throughout the study in order to compare directly to the numerical simulations.

II.D. Transfer Learning from Low-Fidelity Simulations

We adopt deep neural network (DNN) as our ML model for its expressiveness. Our overall goal is then to train a DNN that captures the mapping from design features $x = \{M_c, C_{max}, t_c, P/D, \Omega, U_\infty\}$ to acoustic quantities $y = \{SPL@45^\circ, SPL@90^\circ\}$ (see Figure 4 for the polar angle definitions) exhibited in the high-fidelity experiments:

$$\hat{y} = f(x; w), \quad (1)$$

where $\hat{y}$ denotes the DNN *prediction* of the output quantities, and w collects all tunable DNN parameters (i.e. weight and bias parameters). A depiction of the DNN architecture is shown in Figure 5. Given a training data set $\mathcal{D} = \{x_n, y_n\}_{n=1}^N$, the DNN is trained by finding the optimal w^* that minimizes a loss function $\mathcal{L}$ (e.g., mean squared error):

$$w^* = \arg \min_w \mathcal{L}(w; \mathcal{D}) = \arg \min_w \left\{ \frac{1}{N} \sum_{n=1}^N [f(x_n; w) - y_n]^2 \right\}. \quad (2)$$

However, the expensive experiment data is scarce and may be too few to achieve good training. Therefore we leverage information gained from low-fidelity simulations in order to supplement the high-fidelity data through the process of TL.

The simplest implementation of TL begins by training a model on (typically abundant) data from a *source domain*, in our case that is the low-fidelity simulation data set $\mathcal{D}_{LF}$. This allows us to obtain a DNN whose parameters are

$$w_{LF} = \arg \min_w \mathcal{L}(w; \mathcal{D}_{LF}). \quad (3)$$

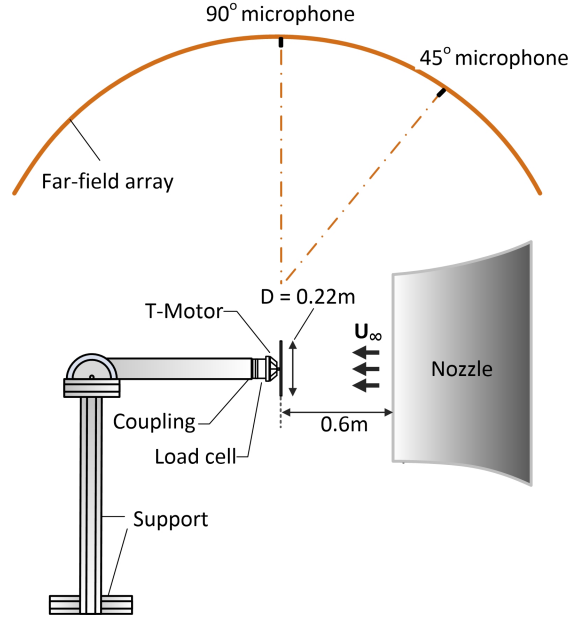


Figure 4: Schematic of the experimental set-up of the propeller noise measurements.

The subsequent task is then to learn modifications to this model in order to adapt the model to a different, but related *target domain* where usually only a small amount of data is available,¹¹ in our case that is the data set of high-fidelity experiments $\mathcal{D}_{\text{HF}}$. We then partition the overall parameters into two subsets $w = \{w_1, w_2\}$, for example where w_2 are the parameters of the last layer of the DNN while w_1 represents the rest. An effective TL technique then entails “freezing” the w_1 to their corresponding values of $w_{\text{LF},1}$, and only retrain w_2 using the new $\mathcal{D}_{\text{HF}}$ (illustrated in Figure 5):

$$w_2^* = \arg \min_{w_2} \mathcal{L}(\{w_{\text{LF},1}, w_2\}; \mathcal{D}_{\text{HF}}). \quad (4)$$

Upon completion, the TL model has final parameters $w = \{w_{\text{LF},1}, w_2^*\}$.

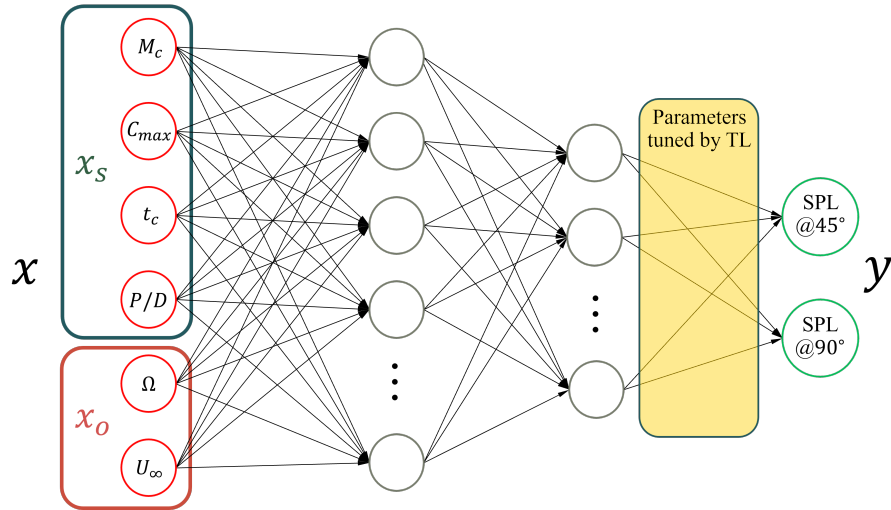


Figure 5: A DNN is first trained using source data set $\mathcal{D}_{\text{LF}}$. During TL, only the last layer’s parameters (highlighted in orange) are retrained using the target data set $\mathcal{D}_{\text{HF}}$ while all other parameters are frozen at values obtained from the source data set. Note that the overall input vector x can be further grouped into shape parameters x_s and operating conditions x_o .

Freezing the parameters from all but the last layer preserves many of the general trends that were learned from the related source domain while still allowing trends unique to the target domain to be learned and corrected. Another

benefit of this TL approach is that the influence of parameter over-fitting is greatly mitigated, since the dimensionality of parameters only from the last layer is substantially smaller than from the entire DNN. Over-fitting is a major concern when learning with small data sets. Training only the last layer of parameters on the target data allows for (or be closer to) an over-determined system, whereas the DNN would otherwise likely be over-parameterized and under-determined if we trained all of its parameters (i.e. from all layers) on the small target data set.

II.E. Active Learning for Physical Experiments

Upon constructing a DNN model via TL using existing data sets, we also seek to further improve the ML model by acquiring high-fidelity data from additional wind-tunnel experiments. As described in Section II.C, the experiments are expensive, therefore it is important that we choose new experiments that can provide the greatest benefit. In the field of ML, the process of determining which input-output pairs to acquire next is called AL. The overall AL cycle for our propeller performance prediction is depicted in Figure 6.

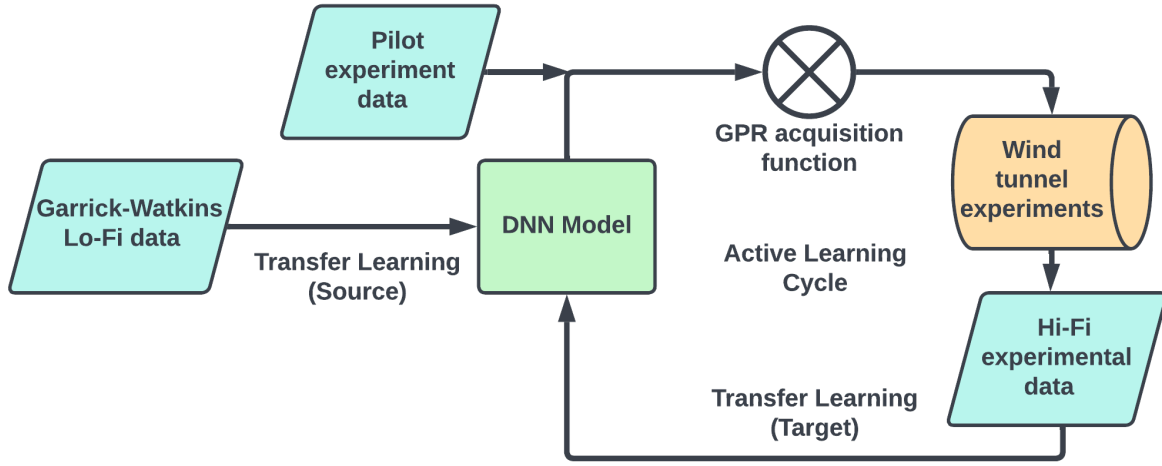


Figure 6: Flow chart showing the iterative process of active learning and transfer learning to improve the neural network model of propeller performance.

More specifically, we wish to propose *only* new shape parameters of propeller blade, since given a shape then the wind-tunnel experiment can sweep through multiple operation conditions at relatively low cost. Thus, we split our overall input vector $x = \{x_s, x_o\}$ into $x_s = \{M_c, C_{max}, t_c, P/D\}$ for the shape parameters and $x_o = \{\Omega, U_\infty\}$ for the operation conditions. Experiments should cover a wide breadth of designs within our design space while also focusing on designs whose performance deviates the most from the low-fidelity simulations (i.e. regions where we currently have high error). This is the classic trade off of exploration versus exploitation in design and optimization. The function used to rank or weigh the value of a future feature-label pair is called an *acquisition function*. While the choice of which acquisition function is often domain specific, we derive our acquisition function by making use of Gaussian process regression (GPR). In particular, we focus on the discrepancy between future experimental data and the existing DNN model

$$\mathcal{E}(x_s) = \mathbb{E}_{x_o} [y - f(x; w)], \quad (5)$$

where f is our DNN prediction from Equation 1, and $\mathbb{E}_{x_o}$ effectively computes the error averaged over the operating condition variables. However, we do not know this discrepancy function exactly (except at locations where we have data points), and so we model its uncertainty as a Gaussian process (GP) (i.e. a Gaussian random field) which is specified by a mean function m and covariance kernel k :

$$\mathcal{E}(x_s) \sim GP(m(x_s), k(x_s, x'_s)). \quad (6)$$

A GP possess the property where any finite set of $\epsilon(x_s^{(1)}), \dots, \epsilon(x_s^{(M)})$ follows a multivariate Gaussian distribution:

$$\begin{bmatrix} \epsilon(x_s^{(1)}) \\ \vdots \\ \epsilon(x_s^{(n)}) \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} m(x_s^{(1)}) \\ \vdots \\ m(x_s^{(n)}) \end{bmatrix}, \begin{bmatrix} k(x_s^{(1)}, x_s^{(1)}) & \dots & k(x_s^{(1)}, x_s^{(n)}) \\ \vdots & \ddots & \vdots \\ k(x_s^{(n)}, x_s^{(1)}) & \dots & k(x_s^{(n)}, x_s^{(n)}) \end{bmatrix} \right). \quad (7)$$

Then, given a set of data points $\mathcal{D}$, we can update the prior GP to a posterior following Bayes' rule, which can be shown to retain a Gaussian form (i.e. remains a GP) but with its new mean ($m(x_s; \mathcal{D})$) and covariance ($k(x_s, x'_s; \mathcal{D})$) that can be easily computed through standard formulas. We use a weighted average of the posterior mean and standard deviation of the posterior GP as our acquisition function to identify the next propeller designs for experimentation.

$$x_{d,new} = \arg \max_{x_s} m(x_s; \mathcal{D}) + 2\sigma(x_s; \mathcal{D}) \quad (8)$$

where $\sigma(x_s; \mathcal{D}) = \sqrt{k(x_s, x_s; \mathcal{D})}$. Such an acquisition function looks for a combination of high mean error from the DNN model as well as high uncertainty, which emerges as a result of being far from existing experimental data points. The mean of error estimate reflects the notion of exploitation (i.e. wanting experiments where we currently have high error), and the standard deviation reflects exploration (i.e. wanting experiments where we are unsure of the behavior).

AL is typically an iterative process of choosing a set of designs (features) at which to collect data, acquiring the corresponding labels for those features, updating the model with the new data, and repeating. Due to the time-intensive process of 3D printing and testing propeller designs, it is not feasible to test a single design before updating the model and identifying the next single design to construct and test. Instead, logistics limit us to testing designs in batches. That is to say, our acquisition function must identify multiple designs to obtain experimental data, before the DNN is updated and the next iteration of new designs are computed. To achieve this, we find the next design, emulate a hypothetical observation of that design and augment it to $\mathcal{D}$, and repeat the process until a batch of desired number of new experiments is proposed.

III. Results

III.A. Low-Fidelity Propeller Performance Prediction and Training Data Generation

The BEMT and Garrick-Watkins (BEMT-GW) simulations were performed on 54,000 propellers inside the design space illustrated in figure 2. This produced the set of low fidelity predictions for the thrust, torque and tonal SPL. The tone at the blade passing frequency was computed for observer locations in the rotor plane (1.75m, 90°) and upstream of the rotor. (1.75m, 45°) (see fig. 4) An example of the tonal directivity predicted by the BEMT-GW simulation is shown in figure 7. The propeller in this case is inside the design space, with a pitch-to-diameter ratio of 1.0 and a NACA 4412 profile at a forward speed of 8.0 m/s and a rotational speed of 5000 RPM. The shape of the directivity pattern of the experimental results are captured by the low-fidelity simulation. The convective amplification of the tone upstream of the rotor is also predicted well by the model. However, there is an under-prediction of the tone by BEMT-GW of around 3-4 dB at most polar angles. This discrepancy is, in part, due to other tonal noise generation mechanisms such as thickness noise not being modelled by the Garrick-Watkins method. This comparison reveals clear limitations of the predictive capabilities of the low-fidelity BEMT-GW solver, and underscores the importance of the proposed data-driven framework in which a machine learning model trained predominantly with low-fidelity results is iteratively enhanced by high-fidelity experimental measurement.

The main advantage of using this method for the generation of the baseline samples is its computational efficiency. The performance of the 54,000 low-fidelity samples used in this study were evaluated in minutes on consumer-grade hardware. In contrast, a higher fidelity simulation of propeller aerodynamics and acoustics such as using an acoustic analogy coupled with unsteady RANS would take many hours on a HPC cluster to evaluate a single case.

III.B. Experiments

Two rounds of experiments were performed with first the preliminary round of 8 randomly chosen blades and the second round a set of 10 blades informed from active learning to improve the prediction accuracy of the DNN. Two more blades were subsequently tested as validation to DNN performance. Table 2 provides the parameter space used in both rounds, including the 2 blades for validation (i.e., randomly chosen from the design space). It is clear from the blade parameter that the active-learning informed parameters tend to incline more towards the higher P/D , max camber, suggesting distinct noise behaviour of the highly cambered blades at high pitch-to-diameter ratios.

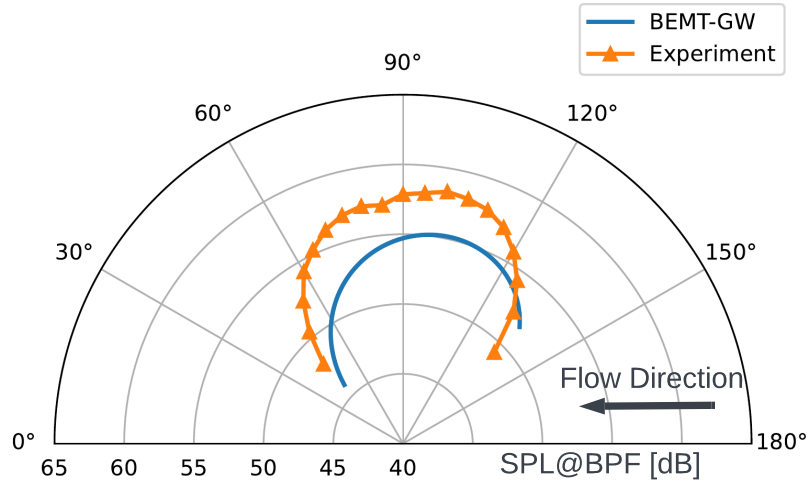


Figure 7: Tonal SPL at the blade passing frequency for the (P/D) 1.0, NACA 4412 propeller at 8.0 m/s, 5000 RPM. Noise measurements from experiment and BEMT are shown on a circular arc with radius 1.75 m. The 90 degree polar angle is aligned with the rotor plane and the flow direction is left to right

Before carrying out detailed analyses and training process for the machine learning models, it is useful to first examine the noise spectra of the in-house designed propellers to validate its performance. Figure 8 shows the noise spectra for propellers with NACA 0012 and NACA 4412 cross-sectional profiles at $P/D = 0.8$ and $P/D = 1.4$, respectively at a free-stream velocity of $U_\infty = 12\text{m/s}$ and rotational speed of 5000rpm. The overall characteristics of the noise spectra consists of a series of tonal peaks at the primary blade pass frequency and the subsequent harmonics. Moreover, traces of elevated noise magnitude, manifested as broadband humps, can be observed at relatively higher frequencies of $f \geq 1000\text{Hz}$. This is typical for a propeller at forward flight conditions.² Comparing between the symmetric NACA 0012 and the cambered NACA 4412, the effect of pitch-to-diameter ratio appears more pronounced for the cambered profile with notable level of broadband noise at $P/D = 1.4$. Moreover, when integrating over the frequency range from $80 \leq f \leq 10000\text{Hz}$, the overall sound pressure level indicates that at lower P/D , NACA 0012 blades produces greater noise than that of the NACA 4412 blade at similar advance ratio (recall that the free-stream velocity and rotational speed is kept constant in the comparison). The opposite is true when the P/D ratio increases from 0.8 to 1.2. Clearly, the effects of the cross-sectional profile of the blade and the pitch-to-diameter ratios are both influential to the noise performance of the propellers, so do the advance ratio (not shown here for the sake of brevity). Hence, these parameters should consist of the key inputs to the training process for the ML model.

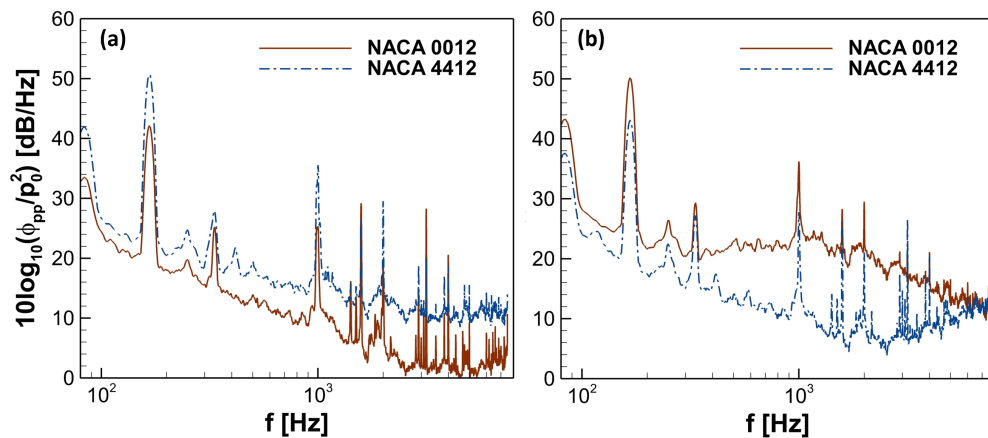


Figure 8: Comparison of the power spectral density of the propeller noise between NACA 0012 and NACA 4412 cross-sectional profiles for (a) $P/D=0.8$ and (b) $P/D=1.4$ at $U_\infty = 12\text{m/s}$ and rotational speed of 5000rpm.

First experimental campaign with randomly selected blades				
	Pitch to diameter ratio (P/D)	Max camber (M_c)	Max camber location (C_{max})	Thickness (t_c)
Blade1A	0.8	0	0	12
Blade1B	1.0	0	0	12
Blade1C	1.2	0	0	12
Blade1D	1.4	0	0	12
Blade1E	0.8	4	4	12
Blade1F	1.0	4	4	12
Blade1G	1.2	4	4	12
Blade1H	1.4	4	4	12
Second experimental campaign with blades informed by active learning				
	Pitch to diameter ratio (P/D)	Max camber (M_c)	Max camber location (C_{max})	Thickness (t_c)
Blade2A	1.4	4	5	12
Blade2B	1.4	5	5	12
Blade2C	1.4	3	5	12
Blade2D	1.4	4	5	10
Blade2E	1.4	4	5	14
Blade2F	1.3	4	5	12
Blade2G	0.9	0.5	2	18
Blade2H	1.0	0.5	2.5	20
Blade2I	0.9	0.5	3.25	20
Blade2J	1.1	1.5	2.0	20
Validation experiments				
	Pitch to diameter ratio (P/D)	Max camber (M_c)	Max camber location (C_{max})	Thickness (t_c)
BladeV1	0.9	3	3	20
BladeV2	1.2	5	4	18

Table 2: Blade shape parameters used for both rounds of experiments and the validation cases (20 in total). Note that all parameters are expressed as percentage to the aerofoil chord of $c = 2.86\text{cm}$.

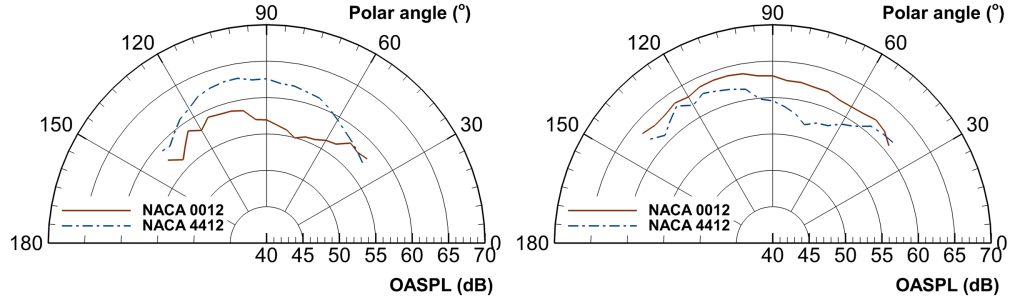


Figure 9: Comparison of the overall sound pressure levels of the propeller noise between NACA 0012 and NACA 4412 cross-sectional profiles for (a) $P/D=0.8$ and (b) $P/D=1.4$ at $U_\infty = 12\text{m/s}$ and rotational speed of 5000rpm.

III.C. Transfer and Active Learning

We adopt a densely connected DNN architecture with two hidden layers, totaling around 1573 tunable parameters (Figure 5 depicts the architecture, and Table 3 presents the number of parameters). All data sets involved are summarized in Table 4 with detailed blade shape parameters for each set captured in Table 2. A baseline DNN with all 1573 tunable parameters is first trained using a data set of 54,000 low-fidelity simulations $\mathcal{D}_{\text{LF}}$. Subsequently, when TL is deployed, parameters are frozen to these low-fidelity values in all except the output layer, and these 85 parameters are retrained using the 106 high-fidelity data points from the pilot set of experiments (and also with the new experiment results from AL once AL is performed). Overall, we will compare 5 different DNN models trained using: (1) low-fidelity data set only (Lo-Fi), (2) pilot high-fidelity data set only without AL data (Hi-Fi 8), (3) high-fidelity data set of both pilot and AL data (Hi-Fi 18), (4) both low- and high-fidelity data sets via TL but without AL data (TL 8), and (5) both low- and high-fidelity data sets via TL including AL data (TL 18).

Layer name	Tunable Parameters (Baseline)	Tunable Parameters (Transfer Learning)
Input	0	0
Dense 1	448	0
Dense 2	1040	0
Output	85	85
Total	1573	85

Table 3: Architecture of the baseline and transfer learned DNN. In the baseline DNN, all layers' parameters are trainable. In the transfer learning model, all but the last layer's parameters are frozen after training on low-fidelity simulations, and the last layer's parameters are retrained based on experimental data.

Data set name	Data Entries	Unique Experimental Geometries
Garrick-Watkins simulations	54,000	n/a
Experimental train: pilot	106	8
Experimental train: AL 1	106	10
Experimental test	23	2

Table 4: Unique experimental geometries are characterized by a unique combination of the shape parameters: $M_c, C_{max}, t_c, P/D$. Data is collected for each shape design at multiple Ω and U_∞ operating conditions.

We carry out AL as follows. Following Section II.E, we use GP to model the error between the DNN trained purely on low-fidelity simulation data and actual performance measured during pilot experiments. The acquisition function identifies the shape parameters that maximized the sum of the GP mean and two standard deviations, averaged across the range of operation condition variables. Once a set of shape parameters is selected, hypothetical performance of that design was simulated and augmented to the existing GP training set. The process of selecting a new set of design specifications is repeated until a batch of 10 new shape designs is determined.

Figure 10 presents the prediction versus experimentally measured SPL values under different design inputs using

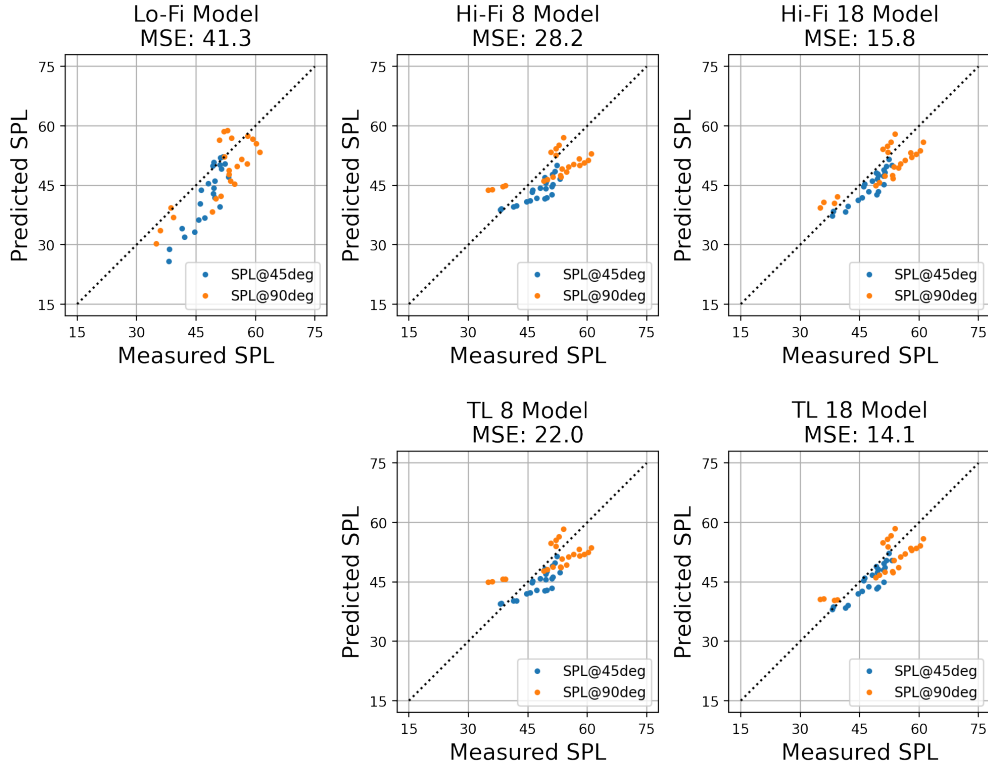


Figure 10: Comparison of predicted versus experimentally measured SPL values of DNN trained using low-fidelity data set (Lo-Fi), high-fidelity data set without (Hi-Fi 8) and with (Hi-Fi 18) AL data, and both low- and high-fidelity data sets via TL without (TL 8) and with (TL 18) AL data. Test MSE values are also shown at the top of each subfigure.

two validation propellers (see bottom of Table 2), where predictions are produced and compared across different DNN models (trained using different combinations of data sets) in the subfigures. If a model makes accurate prediction, the plotted points would be more aligned with the diagonal line in the figures. The overall mean squared error (MSE) is also shown for each ML model in its subfigure. We observe error to be lower when TL is employed than when it is not (comparing top row versus bottom row), and when the additional data from AL is incorporated than when it is not (comparing middle column versus right column). The DNN with the lowest error employed both TL and used the additional AL data set. These observations all support the effectiveness of both TL and AL. Due to time constraints and logistic limitations to wind tunnel testing, we only conduct one iteration of the AL cycle; future work will entail repeating the AL cycle for additional experiments.

IV. Conclusions

This paper presents the first exploratory effort in developing a data-driven framework to improve the prediction of propeller performance. In this framework, a multi-fidelity machine learning (ML) model, based on the transfer learning (TL) and active learning (AL) techniques outlined in Section II.D and II.E, was first trained using a large number of computationally inexpensive low-fidelity simulations and then enhanced by a small number of high-fidelity wind tunnel measurements using TL. To that end, an initial deep neural network was built with 54,000 low-fidelity BEMT-GW training samples. Then the first set of eight propeller blade designs were tested in different operating conditions in the aeroacoustic wind tunnel facility at University of Bristol. The high-fidelity measurement data thus obtained were used to update the initial ML model using a TL approach. To further enhance this multi-fidelity ML model, 10 additional propeller designs were intelligently selected to be manufactured and experimentally measured based on an AL algorithm designed to minimize the predictive error of the ML model. In the deployment stage, this ML model enables rapid prediction of the tonal noise level of isolated 2-bladed propellers at two farfield observer locations given the airfoil profile, pitch-to-diameter ratio as well as forward speed and rotational speed of the propeller.

To demonstrate the efficacy of this data-driven framework, two additional sets of propeller blades were manufac-

tured and tested, and they were held out from ML training. The comparison between these high-fidelity test samples and the ML predictions indicated that the multi-fidelity ML model trained with both low- and high-fidelity data based on the TL approach delivered more accurate predictions than the ML models trained solely with low- or high-fidelity data. Furthermore, the inclusion of additional high-fidelity data set selected by an AL algorithm was shown to further improve the predictive accuracy of the ML model.

The current work focuses on tonal noise prediction of isolated propellers at the blade passing frequency. For future work, we will introduce a trailing-edge noise model in the low-fidelity noise simulation suite to enable broadband noise prediction. The overall multi-fidelity ML model will be extended to correct not only the tonal but also the broadband noise level of isolated propellers, using high-fidelity experimental data. Due to time and logistical constraints, current work also involves only one AL iteration in updating the ML model; future work will entail introducing additional experimental data by performing more AL iterations, thus allowing further improvements to the predictive accuracy of the ML model.

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